```r
# 1. Load Libraries
library(tidyverse)
library(forecast)
library(randomForest)

# 2. Load Data
df_raw <- read.csv(file.choose())

# 3. Smart Cleaning Function
clean_it <- function(x) as.numeric(gsub("[^0-9.-]", "", as.character(x)))

# Find columns by keyword automatically
col_land <- grep("Land", names(df_raw), ignore.case = TRUE)[1]
col_co2 <- grep("Co2", names(df_raw), ignore.case = TRUE)[1]
col_pop <- grep("Population", names(df_raw), ignore.case = TRUE)[1]
col_gdp <- grep("GDP", names(df_raw), ignore.case = TRUE)[1]

# Build the final dataset
df_final <- data.frame(
  GDP    = clean_it(df_raw[, col_gdp]),
  Pop    = clean_it(df_raw[, col_pop]),
  Land   = clean_it(df_raw[, col_land]),
  CO2    = clean_it(df_raw[, col_co2])
)

# Create the Tableau calculation
df_final$GDP_per_capita <- df_final$GDP / df_final$Pop

# Critical: Remove NAs and check if data exists
df_final <- df_final[complete.cases(df_final), ]

if (nrow(df_final) > 0) {
  # --- PART A: RANDOM FOREST ---
  rf_model <- randomForest(GDP_per_capita ~ CO2 + Pop + Land, data = df_final, ntree = 500)
  print(importance(rf_model))

  # --- PART B: ARIMA ---
  # Only use top 20 if we have at least 20 rows
  n_rows <- min(20, nrow(df_final))
  gdp_values <- sort(df_final$GDP, decreasing = TRUE)[1:n_rows]
  gdp_ts <- ts(gdp_values, start = 2003, frequency = 1)

  fit_arima <- auto.arima(gdp_ts)
  gdp_forecast <- forecast(fit_arima, h = 5)

  print(autoplot(gdp_forecast) +
          labs(title = "5-Year GDP Forecast", subtitle = "Top-tier Global Trend Simulation") +
          theme_minimal())
} else {
  print("Error: No data left after cleaning. Check if your CSV columns contain numbers.")
}

# 1. Create a simple data frame from your Random Forest results
imp_data <- as.data.frame(importance(rf_model))
imp_data$Variable <- rownames(imp_data)

# 2. Plot the Importance using ggplot2
ggplot(imp_data, aes(x = reorder(Variable, IncNodePurity), y = IncNodePurity)) +
  geom_col(fill = "steelblue") +
  coord_flip() +  # Makes it a horizontal bar chart for better reading
  labs(
    title = "Key Drivers of GDP per Capita",
    subtitle = "Based on Random Forest 'IncNodePurity' Scores",
    x = "Predictor Variable",
```

```
    y = "Importance (Node Purity)"
  ) +
  theme_minimal()



# Load required libraries
library(tidyverse)

# 2. Load Data
df_raw <- read.csv(file.choose())


df_clean <- df %>%
  mutate(
    # Clean character strings into numbers
    GDP_Numeric = as.numeric(gsub("[$,]", "", GDP)),
    Population_Numeric = as.numeric(gsub(",", "", Population)),
    # Replicating your Tableau Calculated Field
    GDP_per_Capita = GDP_Numeric / Population_Numeric
  ) %>%
  # Filter out rows where GDP or Population is missing to ensure a clean rank
  filter(!is.na(GDP_per_Capita))

# 2. Filter for Top 20 countries BY GDP PER CAPITA
top_20_per_capita <- df_clean %>%
  arrange(desc(GDP_per_Capita)) %>%
  slice(1:20)

# 3. Generate the Bar Chart
ggplot(top_20_per_capita, aes(x = reorder(Country, GDP_per_Capita), y = GDP_per_Capita)) +
  geom_bar(stat = "identity", fill = "darkgreen") +
  coord_flip() +
  scale_y_continuous(labels = scales::dollar_format()) + # Formats axis as currency
  labs(
    title = "Top 20 Countries by GDP per Capita (2023)",
    subtitle = "Ranked by highest economic output per person",
    x = "Country",
    y = "GDP per Capita (USD)"
  ) +
  theme_minimal()



install.packages("plotly")


# Load required libraries
library(tidyverse)
library(plotly)

# 2. Load Data
df_raw <- read.csv(file.choose())

df_clean <- df_raw %>%
  mutate(
    GDP_Numeric = as.numeric(gsub("[$,]", "", GDP)),
    Pop_Numeric = as.numeric(gsub(",", "", Population)),
    CO2_Numeric = as.numeric(gsub(",", "", Co2.Emissions)),
    GDP_per_Capita = GDP_Numeric / Pop_Numeric,
    CO2_per_Capita = CO2_Numeric / Pop_Numeric
  ) %>%
  filter(!is.na(GDP_per_Capita), !is.na(CO2_per_Capita))

# 2. Build the ggplot
```

```
# We add a 'text' aesthetic so Plotly knows what to show on hover
p <- ggplot(df_clean, aes(x = GDP_per_Capita,
                          y = CO2_per_Capita,
                          text = paste("Country:", Country,
                                       "<br>GDP per Cap: $", round(GDP_per_Capita, 0),
                                       "<br>Pop:", format(Pop_Numeric, big.mark=",")))) +
  geom_point(aes(size = Pop_Numeric), alpha = 0.5, color = "midnightblue") +
  geom_smooth(method = "lm", color = "darkorange", fill = "orange", alpha = 0.2) +
  scale_x_log10(labels = scales::dollar_format()) +
  scale_y_log10() +
  labs(title = "CO2 Emissions vs. GDP per Capita",
       x = "GDP per Capita (USD)",
       y = "CO2 Emissions per Capita") +
  theme_minimal()

# 3. Convert to interactive plot
# tooltip = "text" ensures it only shows our custom label on hover
ggplotly(p, tooltip = "text")


# Load libraries
library(tidyverse)
library(plotly)

# 1. Load Data
df_raw <- read.csv(file.choose())

# 2. Clean Data
df_clean <- df_raw %>%
  mutate(
    GDP_Numeric = as.numeric(gsub("[$,]", "", GDP)),
    Pop_Numeric = as.numeric(gsub(",", "", Population)),
    CO2_Numeric = as.numeric(gsub(",", "", Co2.Emissions)),
    GDP_per_Capita = GDP_Numeric / Pop_Numeric,
    CO2_per_Capita = CO2_Numeric / Pop_Numeric
  ) %>%
  filter(!is.na(GDP_per_Capita), !is.na(CO2_per_Capita))

# 3. Build ggplot with trendline fix
# Note: 'text' is only in geom_point to prevent ggplotly from hiding geom_smooth
p <- ggplot(df_clean, aes(x = GDP_per_Capita, y = CO2_per_Capita)) +
  # Points sized by Population and colored by Country (as a proxy for grouping)
  geom_point(aes(size = Pop_Numeric, color = Country,
                 text = paste("Country:", Country,
                              "<br>GDP per Cap: $", round(GDP_per_Capita, 0),
                              "<br>Pop:", format(Pop_Numeric, big.mark=",")),
             alpha = 0.6) +

  # Linear trendline with shading (se = TRUE)
  geom_smooth(method = "lm", color = "black", size = 0.5, se = TRUE, fill = "gray80") +

  scale_x_log10(labels = scales::dollar_format()) +
  scale_y_log10() +
  labs(title = "Global CO2 Emissions vs. Wealth (2023)",
       subtitle = "Hover for details; Trendline shows global correlation",
       x = "GDP per Capita (USD)",
       y = "CO2 Emissions per Capita") +
  theme_minimal() +
  theme(legend.position = "none") # Hiding long country legend for clarity

# 4. Interactive Output
ggplotly(p, tooltip = "text")
```